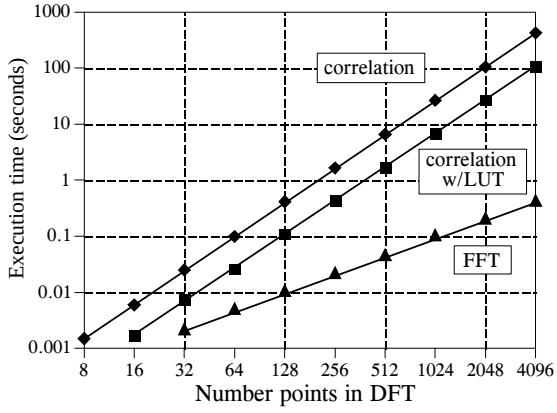

FIGURE 12-8
Execution times for calculating the DFT. The *correlation* method refers to the algorithm described in Table 12-2. This method can be made faster by precalculating the sine and cosine values and storing them in a look-up table (LUT). The FFT (Table 12-3) is the fastest algorithm when the DFT is greater than 16 points long. The times shown are for a Pentium processor at 100 MHz.

The FFT has another advantage besides raw speed. The FFT is calculated more *precisely* because the fewer number of calculations results in less round-off error. This can be demonstrated by taking the FFT of an arbitrary signal, and then running the frequency spectrum through an Inverse FFT. This reconstructs the original time domain signal, *except* for the addition of round-off noise from the calculations. A single number characterizing this noise can be obtained by calculating the standard deviation of the difference between the two signals. For comparison, this same procedure can be repeated using a DFT calculated by correlation, and a corresponding Inverse DFT. How does the round-off noise of the FFT compare to the DFT by correlation? See for yourself in Fig. 12-9.

Further Speed Increases

There are several techniques for making the FFT even faster; however, the improvements are only about 20-40%. In one of these methods, the time

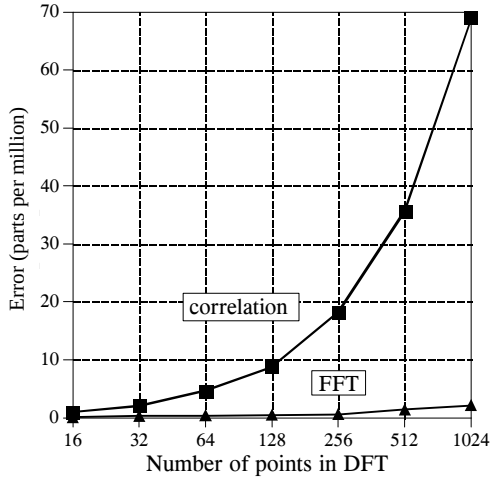

FIGURE 12-9
DFT precision. Since the FFT calculates the DFT *faster* than the correlation method, it also calculates it with less round-off error.